AI JOB SEARCH PLATFORM

ResumeFlow AI

Product Requirements Document — MVP Specification

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PRODUCT	ResumeFlow AI
TYPE	AI-Powered Job Search Platform (SaaS)
STACK	React 19 · TypeScript · Express · MongoDB · OpenAI/OpenRouter · Cloudinary
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Executive Summary

ResumeFlow AI transforms the fragmented job search experience into a single, intelligent platform. Job seekers today juggle separate tools for resume building, AI tailoring, application tracking, and recruiter engagement — ResumeFlow AI unifies all four. By combining an AI-powered resume engine, a Kanban application tracker, a cover letter generator, and real-time recruiter analytics, the platform creates a sticky daily-use product that competes directly with Teal, Enhancv, and Kickresume — while outperforming each of them individually.

Strategic Positioning

Most competitors solve one part of the job search problem well. ResumeFlow AI is the only platform that combines Resume Builder + AI Optimizer + Job Tracker + Recruiter Analytics in a single cohesive workflow.

Core User Journey

The platform is designed around a single, end-to-end job search workflow. Every feature reinforces the next, driving daily retention.

1	**Register / Login**	Auth.js — credentials + OAuth (Google, GitHub)
2	**Dashboard**	Central hub — metrics at a glance
3	**Upload CV**	PDF or DOCX, stored via Cloudinary
4	**AI Parse CV**	Extract structured data from raw CV
5	**Build Master Resume**	Editable, versioned source-of-truth resume
6	**Analyze Job Description**	AI extracts required skills & keywords
7	**Generate Resume Version**	Tailored resume + ATS match score
8	**Generate Cover Letter**	Company & role-specific, AI-written
9	**Apply & Track**	Log application, set status & deadlines
10	**Recruiter Analytics**	See when & how often your resume is viewed

MVP Feature Breakdown

Authentication

- Auth.js (formerly NextAuth.js) — framework-agnostic auth library for Express + React
- Supports: Email/password credentials provider + OAuth (Google, GitHub) out of the box
- Session management via database-backed sessions (MongoDB adapter) or JWT strategy
- Protected route middleware using Auth.js session validation on every API request
- User profile with avatar — pulled from OAuth provider or uploaded via Cloudinary

CV Upload & AI Parsing

- Upload PDF or DOCX — stored on Cloudinary
- AI extracts: contact info, summary, experience, skills, education, projects
- Structured data mapped to MongoDB resume schema
- Handles varied CV formats and layouts gracefully

Master Resume Builder

- Fully editable sections: summary, experience, skills, education, projects, achievements
- Rich inline editing with section reordering
- Resume versioning — every tailored version saved separately
- Single master resume as the source of truth for all versions

AI Resume Tailoring

- Paste job description → AI analyzes required skills & keywords
- AI generates a tailored resume version optimized for the role
- ATS match score shown (0–100%) with specific improvement suggestions
- Multiple templates available for different industries

PDF Export

- Export any resume version as a polished, ATS-friendly PDF
- Template-aware rendering — layout matches selected resume template
- Downloadable and shareable via unique public tracking link

Cover Letter Generator

- AI generates a personalized cover letter per job application
- Inputs: resume version + job description + company name
- Editable before saving or sending
- Stored and linked to the corresponding application

Application Tracker

- Kanban board view: Saved → Applied → Screening → Interview → Technical Assessment → Offer / Rejected
- Table view for sorting and filtering applications
- Each application links to: company, role, resume version, cover letter, status, notes, deadlines
- Visual pipeline overview on the dashboard

Recruiter Analytics

- Unique shareable tracking link per resume version (e.g., resumeflow.ai/r/ahmed-google-fe)
- Events tracked: opens, downloads, unique visitors, timestamps
- Dashboard shows: 'Google recruiter viewed resume 3 times · Last seen 2 days ago'
- IP & user-agent captured per event for deduplication
- Premium differentiator — most competitors lack this entirely

Application Pages

Public Pages

ROUTE	PAGE	DESCRIPTION
`/`	**Landing Page**	Hero, Features, Resume Examples, CTA — conversion-focused public entry point
`/login`	**Login**	Auth.js sign-in page — credentials or OAuth (Google, GitHub)
`/register`	**Register**	Auth.js sign-up with email/password or one-click OAuth

Protected Pages (Authenticated)

ROUTE	PAGE	DESCRIPTION
`/dashboard`	**Dashboard**	Widgets: total resumes, versions, applications, interviews, offers, recruiter views
`/resume/upload`	**CV Upload**	Drag-and-drop PDF/DOCX upload with AI parsing trigger
`/resume/:id`	**Master Resume Builder**	Full resume editor with section management and version history
`/resume/:id/generate`	**Generate Resume Version**	Job description input → AI tailoring → match score → generated version
`/resume/version/:id`	**Resume Version Detail**	View, edit, export PDF, generate cover letter, create tracking link
`/resume/version/:id/analytics`	**Recruiter Analytics**	Total opens, downloads, unique viewers, event timeline
`/applications`	**Application Tracker**	Kanban and table view of all job applications
`/applications/:id`	**Application Detail**	Company, role, status, linked resume version, cover letter, notes, deadlines
`/cover-letters`	**Cover Letters**	List of all generated cover letters with quick access
`/settings`	**Settings**	Profile preferences, account management, notification settings

REST API Endpoints

Authentication

POST	`/api/auth/register`	Auth.js credentials signup — hashes password, creates user
POST	`/api/auth/login`	Auth.js credentials sign-in — returns session token
POST	`/api/auth/logout`	Destroy Auth.js session
GET	`/api/auth/session`	Return current Auth.js session (user + token)

GET	`/api/auth/callback/:provider`	OAuth callback handler (Google, GitHub)

Resumes

POST	`/api/resumes`	Create a new master resume
GET	`/api/resumes`	List all resumes for current user
GET	`/api/resumes/:id`	Get a single resume by ID
PUT	`/api/resumes/:id`	Update resume content
DELETE	`/api/resumes/:id`	Delete resume and all versions

Resume Versions

POST	`/api/versions`	Create a new tailored resume version
GET	`/api/versions/:id`	Get a specific version
PUT	`/api/versions/:id`	Update a version
GET	`/api/resumes/:id/versions`	List all versions for a resume

AI Services

POST	`/api/ai/parse-cv`	Extract structured data from uploaded CV
POST	`/api/ai/analyze-job`	Analyze job description for keywords & requirements
POST	`/api/ai/generate-version`	Generate tailored resume with match score
POST	`/api/ai/improve-section`	Suggest improvements for a specific section
POST	`/api/ai/generate-cover-letter`	Generate personalized cover letter

Applications

POST	`/api/applications`	Create new application record
GET	`/api/applications`	List all applications
GET	`/api/applications/:id`	Get single application detail
PUT	`/api/applications/:id`	Update status, notes, deadlines
DELETE	`/api/applications/:id`	Remove application

Analytics

GET	`/api/analytics/:resumeVersionId`	Aggregate analytics for a version
POST	`/api/analytics/track-view`	Log recruiter view event

POST	`/api/analytics/track-download`	Log recruiter download event

Export

POST	`/api/export/pdf`	Generate and return PDF for a resume version

Database Schema — MongoDB Collections

users

FIELD	TYPE	NOTES
`_id`	ObjectId	Auto-generated primary key
`name`	String	Full display name
`email`	String	Unique, indexed, lowercase
`passwordHash`	String	Auth.js hashed password — only set for credentials provider
`emailVerified`	Date	Set by Auth.js on email verification — null until verified
`provider`	String	Auth provider: 'credentials' | 'google' | 'github'
`avatar`	String	Cloudinary URL
`createdAt`	Date	Account creation timestamp

resumes

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key
`userId`	ObjectId	Ref → users
`title`	String	User-defined resume label
`contact`	Object	Name, email, phone, location, LinkedIn, GitHub
`summary`	String	Professional summary paragraph
`experiences`	Array	Each: title, company, dates, bullets[]
`projects`	Array	Each: name, description, techStack[], url
`skills`	Array	Grouped skill categories
`education`	Array	Degree, institution, dates, GPA
`achievements`	Array	Awards, certifications, publications
`createdAt`	Date	Creation timestamp

resume_versions

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key
`userId`	ObjectId	Ref → users
`resumeId`	ObjectId	Ref → resumes (parent)
`versionName`	String	e.g., 'Google SWE — June 2025'
`companyName`	String	Target company name
`jobTitle`	String	Target job title

`jobDescription`	String	Raw job description text
`template`	String	Selected template identifier
`matchScore`	Number	ATS match score (0–100)
`generatedResume`	Object	AI-tailored resume content
`publicLinkSlug`	String	Unique slug for tracking link — indexed
`createdAt`	Date	Generation timestamp

applications

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key
`userId`	ObjectId	Ref → users
`companyName`	String	Company applied to
`jobTitle`	String	Role applied for
`resumeVersionId`	ObjectId	Ref → resume_versions
`coverLetterId`	ObjectId	Ref → cover_letters (nullable)
`status`	String	Enum: Saved | Applied | Screening | Interview | Technical | Offer | Rejected
`appliedDate`	Date	When application was submitted
`notes`	String	Free-text notes
`deadlines`	Array	Interview dates, task deadlines
`createdAt`	Date	Record creation timestamp

cover_letters

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key
`userId`	ObjectId	Ref → users
`resumeVersionId`	ObjectId	Ref → resume_versions
`companyName`	String	Target company
`jobTitle`	String	Target role
`content`	String	Full cover letter text (editable)
`createdAt`	Date	Generation timestamp

recruiter_analytics

FIELD	TYPE	NOTES
`_id`	ObjectId	Primary key
`resumeVersionId`	ObjectId	Ref → resume_versions — indexed
`eventType`	String	Enum: view | download

`ipAddress`	String	Hashed for privacy
`userAgent`	String	Browser/device fingerprint
`createdAt`	Date	Event timestamp

Project Structure — Monorepo

A monorepo structure is strongly recommended. It keeps the client, server, and any shared types/utilities in a single repository, simplifying deployment, type sharing, and CI/CD pipelines.

client/src/	server/src/
`components/ pages/ hooks/ services/ store/ layouts/ lib/ types/`	`controllers/ routes/ services/ middlewares/ models/ utils/ config/ prompts/`

Tech Stack

Frontend

- React 19 — latest concurrent rendering features
- TypeScript — full type safety across client and server
- Tailwind CSS v4 — utility-first with CSS variable theming
- shadcn/ui — accessible, unstyled component primitives
- Light / Dark / System theme support

Backend

- Express.js (TypeScript) — clean MVC-style REST API
- Mongoose — typed MongoDB ODM with schema validation
- Auth.js — credentials + OAuth providers, MongoDB session adapter
- Multer + Cloudinary — file uploads with cloud CDN delivery

AI Layer

- OpenRouter or OpenAI — switchable via environment config
- Dedicated prompt files per AI action (parse, tailor, cover letter, analyze)
- Structured JSON output enforced for all AI responses
- Token usage tracked per user for future rate limiting / billing

DevOps & Infrastructure

- MongoDB Atlas — managed cloud database with automated backups
- Cloudinary — resume file and avatar storage CDN
- Environment-based config (.env) for all secrets
- Recommended deploy: Railway / Render (API) + Vercel (client)

Development Roadmap — MVP Phases

Phase 1 — Foundation

- ✓ Project scaffolding: monorepo, ESLint, Prettier, TypeScript config
- ✓ Express server setup with middleware (CORS, helmet, morgan, error handler)
- ✓ MongoDB connection, Mongoose models for users and resumes
- ✓ Auth.js setup: credentials provider + Google/GitHub OAuth, MongoDB adapter for sessions
- ✓ CV upload endpoint with Cloudinary integration

Phase 2 — Resume Core

- AI CV parsing service (OpenAI/OpenRouter) with structured JSON output
- Master resume builder API (CRUD) with Mongoose schema
- Resume versioning: generate tailored version from job description
- ATS match scoring returned alongside generated version
- PDF export endpoint using headless rendering

Phase 3 — Job Search Tools

- Cover letter generator API (linked to resume version + job description)
- Application tracker CRUD with status state machine
- Kanban and table view data endpoints
- Application linked to resume version and cover letter

Phase 4 — Analytics & Dashboard

- Recruiter tracking link generation (unique slug per version)
- Track view and download events with IP + user agent
- Analytics aggregation endpoint (totals, unique visitors, timeline)
- Dashboard metrics API: resume count, application funnel, recruiter views

Recruiter Analytics — Deep Dive

This is the platform's most powerful premium differentiator. When a user shares their resume via a tracking link (e.g., resumeflow.ai/r/ahmed-google-fe), every recruiter interaction is silently logged. The user sees a live feed of recruiter activity on their analytics dashboard:

EVENT	DETAIL	TIMESTAMP
■ View	Google · Chrome · macOS	2 days ago
■ View	Google · Chrome · macOS	2 days ago
■ Download	Google · Safari · iPhone	1 day ago
■ View	Google · Firefox · Windows	6 hours ago

Example: 'Google recruiter viewed your resume 3 times. Last seen 6 hours ago.'

Environment Configuration

MongoDB

```
MONGO_URI
```

Auth.js

```
AUTH_SECRET
AUTH_GOOGLE_ID
AUTH_GOOGLE_SECRET
AUTH_GITHUB_ID
AUTH_GITHUB_SECRET
```

Cloudinary

```
CLOUDINARY_CLOUD_NAME
CLOUDINARY_API_KEY
CLOUDINARY_API_SECRET
```

AI

```
AI_API_KEY
AI_BASE_URL
```

App

```
CLIENT_URL
PORT=5000
NODE_ENV=development
```